ASSIGNMENT – 1 FINAL REPORT

Food Delivery Time Prediction

Project Overview

Objective :

The objective of this project is to predict food delivery time based on various factors such as:

- Distance
- Weather conditions
- Traffic conditions
- Vehicle type
- Delivery personnel experience
- Order cost and ratings

Two models were implemented:

- **Linear Regression** (for predicting delivery time)
- **Logistic Regression** (for classifying deliveries as Fast or Delayed)

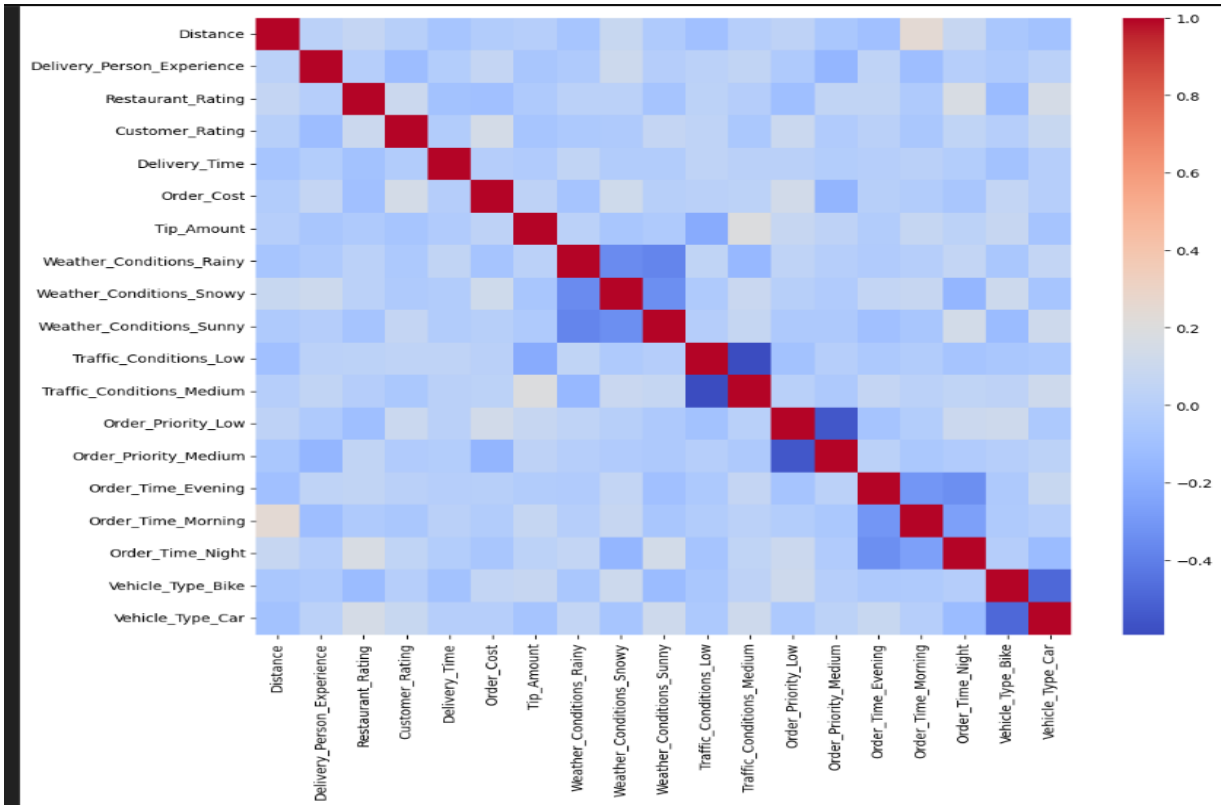
Data Preprocessing

- Removed irrelevant columns (Order_ID, raw latitude & longitude)
- Encoded categorical variables using One-Hot Encoding
- Standardized numeric features using StandardScaler
- Created a new feature: **Rush_Hour**

Exploratory Data Analysis (EDA)

3.1 Correlation Analysis

The correlation heatmap below shows relationships between numerical features and encoded categorical variables.

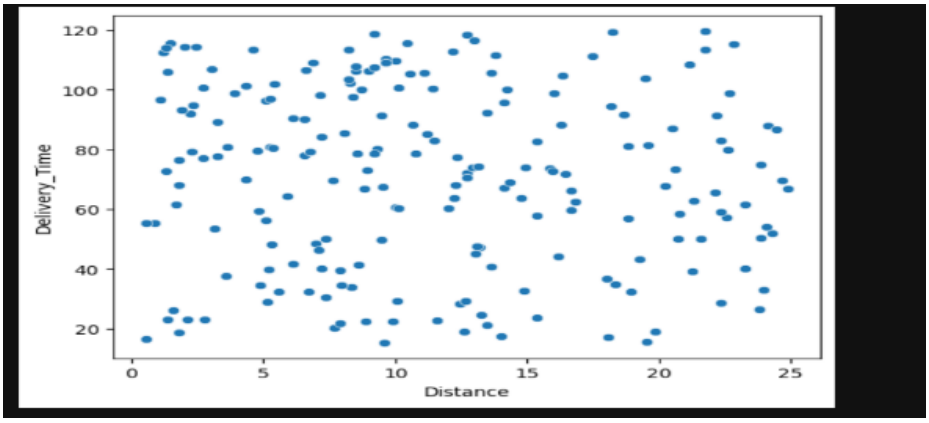


Interpretation:

- Most features show very weak correlation with Delivery_Time.
- No strong linear predictors were observed.
- Encoded categorical variables show expected internal correlations.
- No severe multicollinearity detected.

This indicates that predicting delivery time using simple linear models may be challenging.

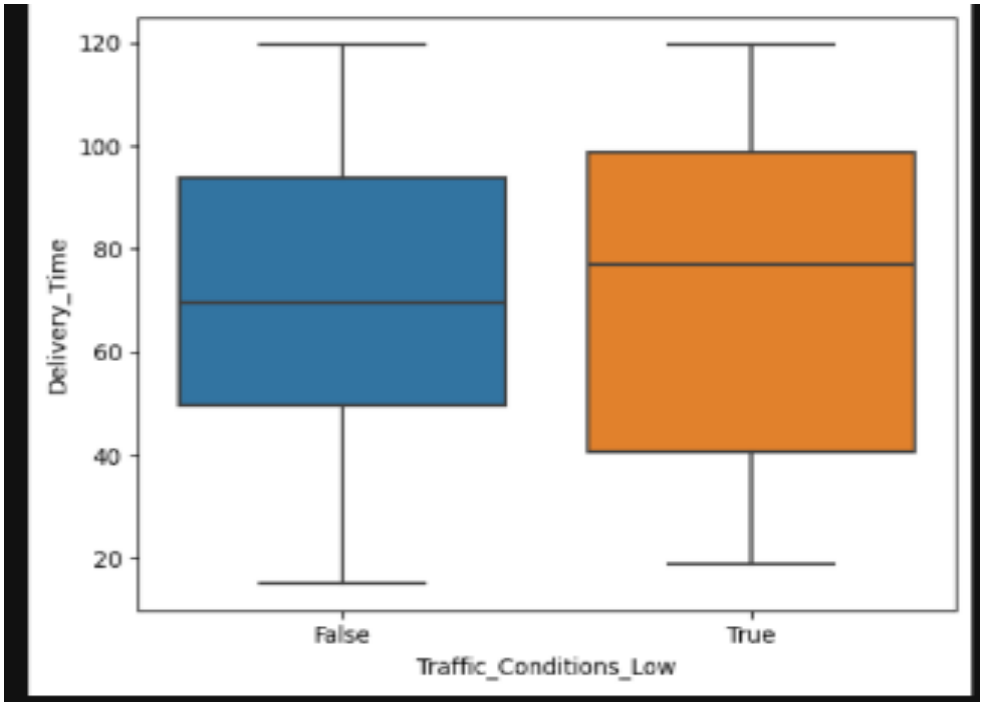
3.2 Distance vs Delivery Time



Interpretation:

- Points are randomly scattered.
- No clear upward or downward trend.
- Indicates weak linear relationship between Distance and Delivery_Time.

3.3 Traffic Effect on Delivery Time



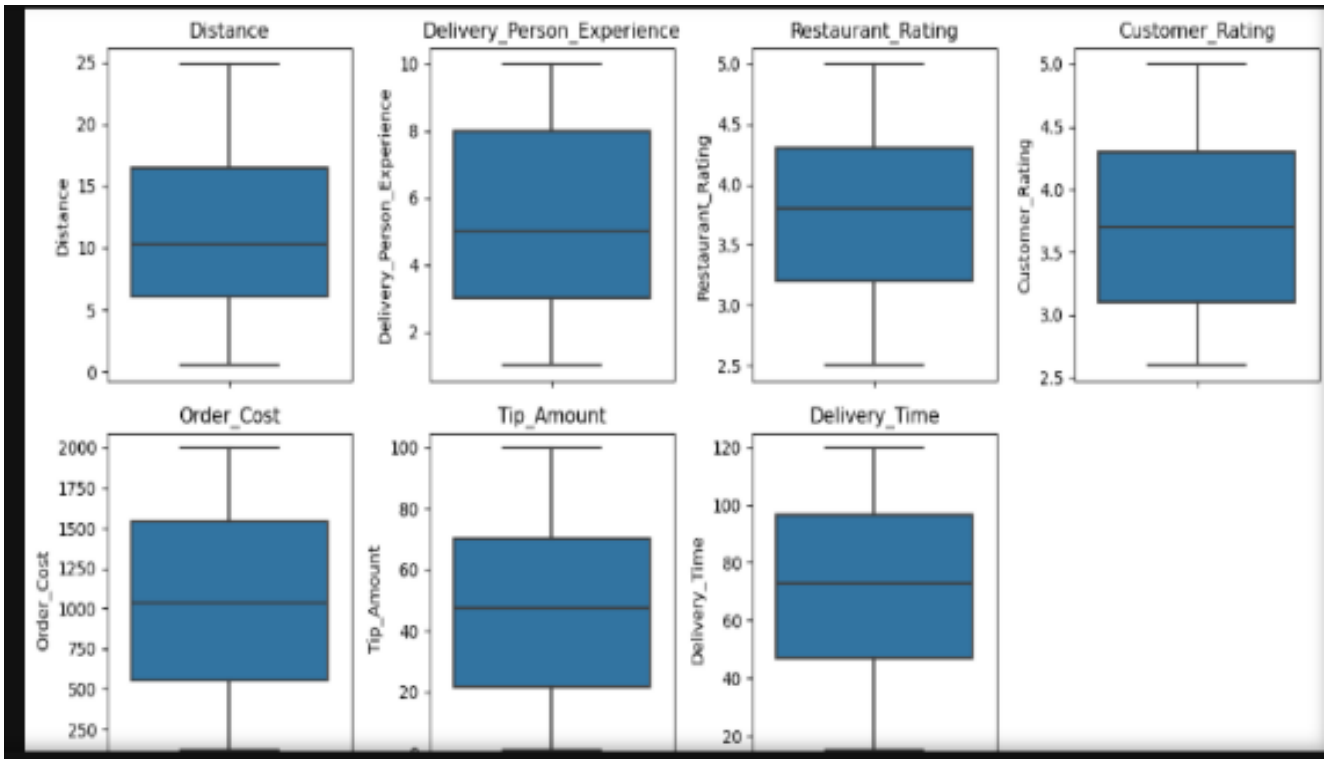
Interpretation:

- Slight variation in median delivery times.
- However, overlap is significant.
- Traffic alone does not strongly determine delivery duration.

3.4 Outlier Detection

Interpretation:

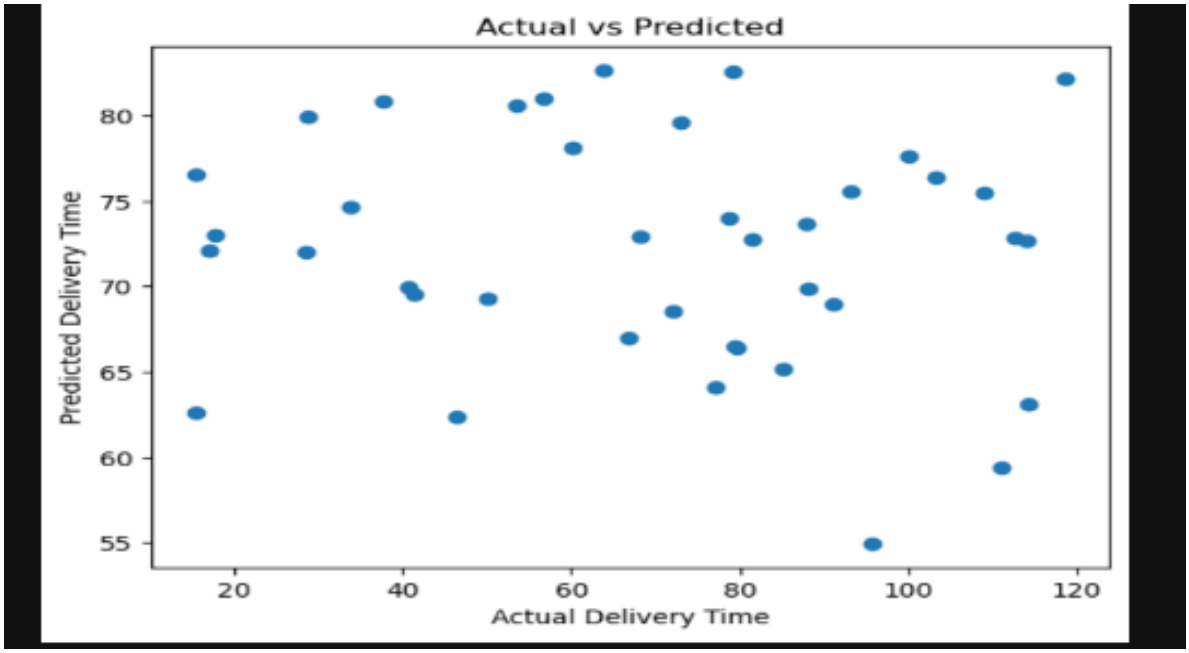
- Some variability in Order_Cost and Delivery_Time.
- No extreme outliers severely distorting the dataset.
- Distribution appears relatively uniform.



Linear Regression Model

Model Evaluation Metrics

- **MSE:** 1021.94
- **MAE:** 27.18 minutes
- **R^2:** -0.10



Interpretation:

- Points do not align along a diagonal.
- Predictions cluster around average values.
- Negative R^2 indicates poor model fit.
- Model performs worse than predicting the mean.

Conclusion:

Linear Regression fails to capture meaningful patterns in the dataset.

Logistic Regression Model

To perform classification, Delivery_Time was converted into:

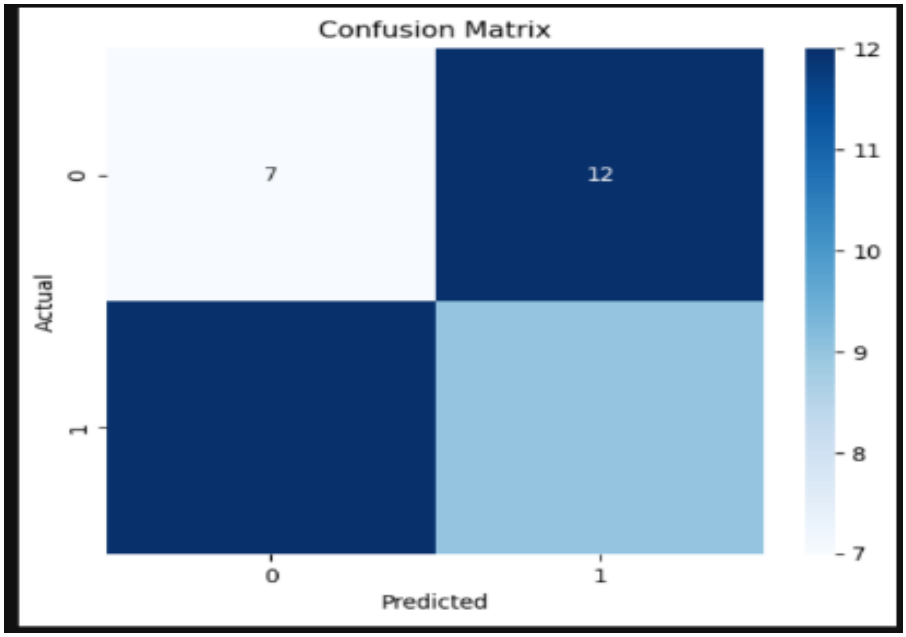
- 0 → Fast
- 1 → Delayed

(based on median split)

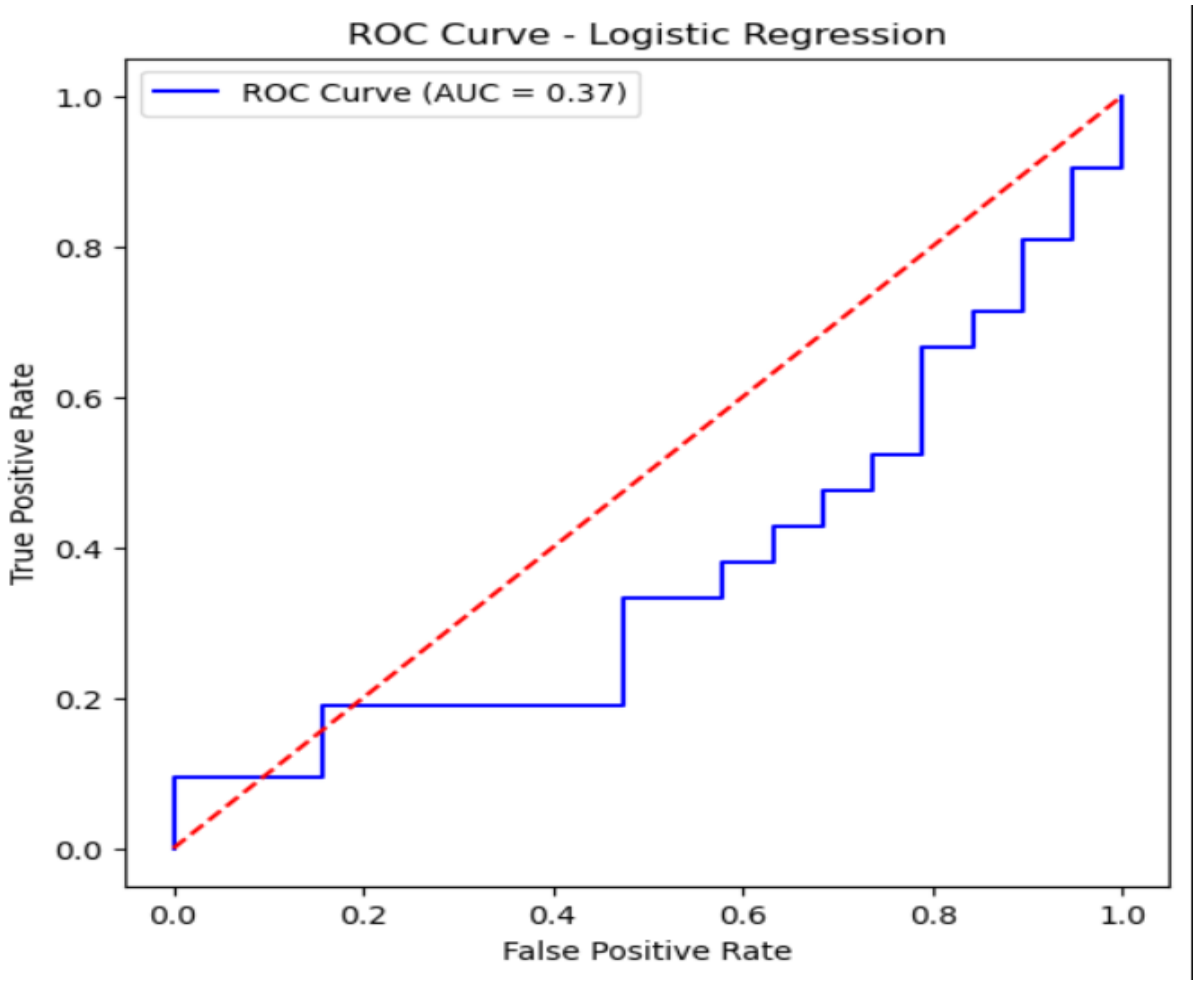
Evaluation Metrics

- **Accuracy:** 40%
- **Precision:** 0.43
- **Recall:** 0.43
- **F1 Score:** 0.43
- **AUC:** 0.37

Confusion Matrix



ROC Curve



Interpretation:

- AUC = 0.37
- Model has poor class separability.
- Performance is below the random baseline (0.5).

Conclusion:

Logistic Regression is unable to effectively classify delivery status.

Model Comparison

Metric	Linear Regression	Logistic Regression
Task	Predict Time	Classify Status
Performance	$R^2 = -0.10$	Accuracy = 40%
Reliability	Poor	Poor

Key Observations

1. Very weak linear relationships exist between features and delivery time.
2. Distance does not strongly influence delivery time in this dataset.
3. Traffic and weather show limited predictive impact.
4. Dataset may contain synthetic or weakly structured data.

Actionable Insights

- Even though predictive accuracy is low, the analysis suggests:
- Improve Traffic Planning
- Increase staffing during rush hours.
- Route Optimization
- Implement shortest-path algorithms for efficient routing.
- Delivery Personnel Training
- Improve efficiency through skill development programs.

Final Conclusion

This project successfully demonstrated the complete machine learning workflow:

- Data preprocessing
- Feature engineering
- Exploratory analysis
- Regression and classification modelling
- Performance evaluation

Although both models performed poorly, the project highlights the critical importance of data quality and meaningful feature selection in predictive modelling.